The Job Sequencing and Tool Switching Problem

Part VII: SSP Variants that Collapse to JGP

Scope and Status. This document addresses the question: can we design a variant of the SSP with a modified cost function f such that solving the variant *collapses* to solving the JGP? We present: (i) a formal definition of collapse; (ii) the TSIP as a known baseline case (**proved in literature**); (iii) **sufficient conditions** for collapse (C1+C2+C3) — **proved**; (iv) the magazine setup time variant S_{stop} and when it causes collapse — **proved**; (v) a **necessity analysis** of the conditions (section 6): (C3) is necessary, while (C1) and (C2) are not, each with an explicit witness. The document explicitly distinguishes proved from open results.

Contents

What Does "Collapse to JGP" Mean?

Definition 1.1 (SSP-f variant)**.** Given a cost function $f : \mathcal{C} \times \mathcal{C} \to \mathbb{R}_{\geq 0}$, the SSP-$f$ variant minimises $\sum_k f(C_k, C_{k+1})$ over all feasible job sequences and magazine configurations.

Definition 1.2 (Collapse to JGP)**.** SSP-f *collapses to JGP* if for every instance I:

$$\mathrm{OPT}_{\text{SSP-}f}(I) \ \text{ is achieved by a solution using exactly } Z^*_{\mathrm{JGP}}(I) \text{ groups.}$$

Equivalently: using more than Z^*_{JGP} groups never strictly reduces the SSP-f objective.

If collapse holds, the JGP can be solved first and the SSP-f objective minimised within those groups, dramatically simplifying the problem.

Baseline: The Tool Switching Instants Problem (TSIP)

Definition 2.1 (TSIP cost function)**.** $f_{01}(C_k, C_{k+1}) = \mathbf{1}[C_k \neq C_{k+1}]$: the cost is 1 if the configuration changes, 0 otherwise.

The TSIP minimises the total number of *configuration changes* (regardless of how many tools are swapped each time). With K groups, there are at most $K - 1$ configuration changes, so the TSIP objective is $\leq K - 1$.

Theorem 2.2 (TSIP collapses to JGP [Adjiashvili et al.(2015)Adjiashvili, Bosio, and Zemmer])**.** *Under f_{01}, the optimal SSP-f_{01} solution uses exactly Z^*_{JGP} groups.*

Proof. Any solution with K configuration changes corresponds to processing K groups (one per distinct configuration used). So $\mathrm{OPT}_{\mathrm{TSIP}} = Z^*_{\mathrm{JGP}} - 1$ (minimum configuration changes = minimum groups minus one). □ □

Remark 2.3. The TSIP is solvable in polynomial time when tool loading/unloading is unconstrained [Adjiashvili et al.(2015)Adjiashvili, Bosio, and Zemmer]. It models situations where the machine must stop completely for each tool change (the "batch" model).

Sufficient Conditions for Collapse (Proved)

We give general conditions under which a cost function f causes SSP-f to collapse to JGP.

Theorem 3.1 (Sufficient conditions for collapse)**.** *SSP-f collapses to JGP if f satisfies all three of the following conditions:*

(C1) Self-cost zero: *$f(C, C) = 0$ for all $C \in \mathcal{C}$.*

(C2) Strict inter-group positivity: *There exists $\delta > 0$ such that for any JGP-optimal grouping $\mathcal{G}^*$, any two groups $G_k \neq G_{k'}$, and any valid configurations $C \in \mathcal{C}(G_k)$, $C' \in \mathcal{C}(G_{k'})$: $f(C, C') \geq \delta > 0$.*

(C3) No-beneficial-split: *For every feasible grouping $\mathcal{G}$ with $K > K^*$ groups, the SSP-f cost for any solution over $\mathcal{G}$ is $\geq$ the SSP-f cost for the best K^*-group solution.*

Proof. Under (C2), any solution with K groups incurs $\geq K-1$ inter-group transitions, each costing $\geq \delta > 0$, so total cost $\geq (K-1)\delta$. For $K > K^*$: $(K-1)\delta > (K^*-1)\delta$, meaning a K^*-group solution with cost $(K^*-1)\delta$ (e.g., by using the TSIP baseline) is strictly cheaper than any $(K+1)$-group solution. Condition (C3) ensures that no splitting of groups into more pieces can further reduce cost. Together, every optimal solution under SSP-f uses exactly K^* groups. □ □

Remark 3.2 (Necessity — partial answer, with an open core)**.** The conjunction (C1)+(C2)+(C3) is *sufficient* but not necessary: the necessity analysis of section 6 (proposition 6.1) shows (C3) is necessary while (C1) and (C2) are each dispensable, via explicit witnesses (e.g. collapse can hold with positive self-cost, violating (C1), or with some zero-cost inter-group transitions, violating (C2)). What remains open is a single *necessary and sufficient* characterisation of collapse; section 6 reduces this to the threshold $\rho_c(I)$ but does not close it.

The Magazine Setup Time Variant (Proved)

Model

In many FMS environments, changing any tool requires stopping the machine. This stop incurs a fixed overhead S_{stop} (machine re-initialisation, operator time) in addition to the per-tool switching cost c_{switch}. The total cost of a transition from C_k to C_{k+1} is:

$$f_S(C_k, C_{k+1}) = S_{\text{stop}} \cdot \mathbf{1}[C_k \neq C_{k+1}] + c_{\text{switch}} \cdot |C_{k+1} \setminus C_k|.$$

When $S_{\text{stop}} = 0$ this reduces to the standard U-SSP. When $c_{\text{switch}} = 0$ (no individual switch cost) this is the TSIP (Section 2). The interesting regime is $S_{\text{stop}} > 0$ and $c_{\text{switch}} > 0$.

Collapse Condition

Theorem 4.1 (Setup-time collapse to JGP)**.** *If the magazine setup time satisfies*

$$S_{\text{stop}} \;>\; n \cdot b \cdot c_{\text{switch}},$$

then SSP-f_S collapses to JGP: the optimal solution uses exactly Z^*_{JGP} groups.

Proof. Consider two competing solutions: one with K groups (configuration changes) and one with $K+1$ groups.

The maximum possible total *per-tool* switching cost across the entire sequence of n jobs and b-capacity magazine is bounded by $n \cdot b \cdot c_{\text{switch}}$ (at most b insertions per job, n jobs total).

So any solution with K groups costs at most $K \cdot S_{\text{stop}} + n \cdot b \cdot c_{\text{switch}}$. Any solution with $K+1$ groups costs at least $(K+1) \cdot S_{\text{stop}}$.

Under the condition $S_{\text{stop}} > n \cdot b \cdot c_{\text{switch}}$:

$$(K+1)S_{\text{stop}} > K \cdot S_{\text{stop}} + n \cdot b \cdot c_{\text{switch}} \;\geq\; \text{cost of any } K\text{-group solution.}$$

Therefore, no $(K+1)$-group solution can beat the best K-group solution. Since this holds for any K, the optimal number of groups is minimised — exactly the JGP objective. □ □

Remark 4.2 (Interpretation)**.** When the setup time is large enough to dominate the per-tool cost of the *entire* sequence, it is always better to minimise stops (groups) than to minimise individual tool insertions. The boundary $S_{\text{stop}} = n \cdot b \cdot c_{\text{switch}}$ is the *collapse threshold*. Below this threshold, the SSP and JGP optima diverge.

Parameterised Collapse: The ρ-Frontier

Status: Partially proved; approximation ratio analysis proposed.

Define the ratio $\rho = S_{\text{stop}}/c_{\text{switch}}$. As ρ increases from 0 to ∞:

- $\rho = 0$: pure U-SSP; JGP and SSP optima can differ (gap analysis in `05_jgp_ssp_gap_analysis.tex`).
- $\rho > nb$: complete collapse (Theorem 4.1).
- $0 < \rho < nb$: partial regime where the optimal number of groups is between Z^*_{JGP} and some $K_\rho > Z^*_{\text{JGP}}$.

Approximation ratio as a function of ρ. Define the JGP+GSP TD-SSP approximation ratio at parameter ρ:

$$R(\rho) = \sup_I \frac{(K^* - 1)\rho + H(I)}{(K^* - 1)\rho + Z^*_{\text{SSP}}(I)},$$

where $H(I)$ is the JGP+GSP cost (U-SSP component only) and $Z^*_{\text{SSP}}(I)$ is the U-SSP optimum. For the 6-job ring (`05_jgp_ssp_gap_analysis.tex`, Example 1), $K^* = 3$, $H = 4$, $Z^*_{\text{SSP}} = 3$:

$$R(\rho)_{\text{ring}} = \frac{2\rho + 4}{2\rho + 3}.$$

At $\rho = 0$: ratio $= 4/3$. At $\rho = 1$: ratio $= 1$ (JGP+GSP is optimal). For $\rho > 1 = \rho_c(\text{ring})$, the JGP+GSP solution is exactly optimal for the TD objective.

Two boundary values are established:

- $R(0) \leq 4/3$ for $b = 3$ (conjectured tight; see `05_jgp_ssp_gap_analysis.tex`, §5).
- $\lim_{\rho\to\infty} R(\rho) = 1$ (JGP+GSP optimal in the TSIP regime).

Practical implication. In CNC manufacturing, the machine stop time S_{stop} is typically 5–30 minutes (magazine swap, re-initialisation) while individual tool change costs c_{switch} are 0.5–2 minutes, giving $\rho \approx 3$–60 [Privault and Finke(1995)]. For ring-like instances ($\rho_c = 1$), all practical parameter values satisfy $\rho \gg \rho_c$: JGP+GSP is exactly optimal for the TD objective. The U-SSP gap (at $\rho = 0$) is therefore largely irrelevant in practice for such instances.

Bi-objective interpretation. The TD-SSP family parameterised by ρ is the weighted-sum scalarisation of:

$$\begin{aligned} &\min\ (K - 1) && \text{(configuration changes = JGP objective)}, \\ &\min\ Z_{\text{switch}} && \text{(tool changes = U-SSP objective)}. \end{aligned}$$

As ρ varies, the optimal solution traces the Pareto frontier. JGP+GSP always achieves the JGP-efficient point $(K^* - 1, H)$. The U-SSP optimal sequence may achieve a different Pareto-efficient point $(K > K^* - 1, Z^*_{\text{SSP}} < H)$. For $\rho > \rho_c(I)$, the JGP point dominates. The multi-objective structure is studied for related tool-switching variants in [Furrer and Mütze(2017)].

Open problem. Characterise the exact threshold ρ_c per instance, and the function $R(\rho)$ as a closed form. This is also related to the Lagrangian relaxation of `06_grouping_selection.tex`, where $\lambda \approx \rho$ penalises extra groups.

The TD-SSP and Lexicographic Collapse

Tool-Dependent SSP (TD-SSP)

In the *Tool-Dependent SSP* (TD-SSP), the cost of inserting tool t is $c_t > 0$ (not necessarily uniform). The transition cost is:

$$f_{\text{TD}}(C_k, C_{k+1}) = \sum_{t \in C_{k+1} \setminus C_k} c_t.$$

When $c_t = 1$ for all t, this reduces to the U-SSP. When c_t are arbitrary, the KTNS policy is no longer optimal (it assumes uniform costs). However, the Tooling Problem (TP) for a fixed sequence remains polynomial via minimum-cost network flow on an interval graph (totally unimodular constraint matrix) [Privault and Finke(1995)].

Lexicographic Collapse Under Heavy Penalties

When a machine stop incurs a *fixed* penalty S_{stop} in the TD-SSP model, the total cost is:

$$Z_{\text{TD}} = K \cdot S_{\text{stop}} + \sum_{k=1}^{K-1} f_{\text{TD}}(C_k, C_{k+1}).$$

Theorem 5.1 (Lexicographic JGP recovery for TD-SSP)**.** *If $S_{\text{stop}} > n \cdot b \cdot \max_t c_t$, the TD-SSP objective is* lexicographically*: first minimise the number of groups K (JGP objective), then minimise the total tool-switching cost within those groups.*

Proof. The maximum possible total tool-switching cost (second objective) across the entire sequence is bounded by $\sum_k \sum_{t \in C_{k+1} \setminus C_k} c_t \leq n \cdot b \cdot \max_t c_t$. Under the stated condition, this entire sum is strictly less than S_{stop}. Therefore, adding one extra machine stop (increasing K by 1) costs at least S_{stop}, which strictly dominates any possible savings in the per-tool cost. The primary objective is therefore to minimise K (JGP), and the secondary objective (per-tool cost) is resolved within the Z^*_{JGP}-group structure. □ □

Remark 5.2 (Practical relevance)**.** This theorem justifies the use of the JGP as a primary objective in settings where machine downtime is very costly (e.g., aerospace manufacturing, high-precision machining). In such settings, the JGP+GSP heuristic is *exactly optimal* for the modified objective, not merely an approximation.

Analysis of Collapse Conditions

Which Conditions are Necessary?

The conditions (C1)+(C2)+(C3) in Theorem 3.1 are *sufficient* for collapse, but not all are necessary.

Proposition 6.1 (Necessity analysis)**.** *Of the three conditions (C1), (C2), (C3) in Theorem 3.1:*

(i) (C3) *is necessary by definition of collapse.*
(ii) (C1) *is* not *necessary in general.*
(iii) (C2) *is* not *necessary in general.*

Proof. **(C3) necessary.** If there exists a grouping with $K > K^*$ groups whose SSP-f cost is strictly lower than all K^*-group solutions, collapse fails by definition. So (C3) is necessary.

(C1) not necessary. Let $f(C, C') = \varepsilon > 0$ for all pairs including $C = C'$ (positive self-cost). Any solution with K groups pays $(K-1)\varepsilon$ regardless of self-costs. Minimising K still gives collapse, so (C1) fails (self-cost positive) yet collapse holds.

(C2) not necessary. Take $J = \{j_1, j_2, j_3\}$, $b = 3$, $T_1 = \{t_1, t_2, t_3\}$, $T_2 = \{t_3, t_4, t_5\}$,

$T_3 = \{t_1, t_4, t_6\}$. Here $K^* = 3$ (each pair requires > 3 tools). Define: $f(C_1, C_2) = 0$, $f(C_2, C_3) = f(C_1, C_3) = 1$, $f(C, C) = 0$. The best 3-group solution costs $f(C_1, C_2) + f(C_2, C_3) = 0 + 1 = 1$. No 4-group solution achieves cost < 1 (it still must cross the boundary between some pair with positive cost). Collapse holds despite $f(C_1, C_2) = 0$ violating (C2). □ □

Remark 6.2 (Minimal sufficient condition)**.** (C3) is the only necessary condition, but it is tautological (it states the definition of collapse). The conditions (C1)+(C2)+(C3) form a clean and *verifiable* sufficient criterion. Finding a tight non-trivial necessary-and-sufficient characterisation of all f for which SSP-f collapses to JGP remains open.

The Collapse Threshold $\rho_c(I)$

For the setup-time model f_S with ratio $\rho = S_{\text{stop}}/c_{\text{switch}}$, the exact collapse threshold is computable per instance.

Theorem 6.3 (Collapse threshold formula)**.** *Define the* switch-savings function*:*

$$\Delta(I, K) = \min_{K^*\text{-group solutions}} \text{cost}_{c=1}(\cdot) - \min_{K\text{-group solutions}} \text{cost}_{c=1}(\cdot), \quad K > K^*.$$

The collapse threshold is $\rho_c(I) = \sup_{K > K^*} \Delta(I, K)/(K - K^*)$. SSP-$f_S$ collapses to JGP if and only if $\rho > \rho_c(I)$.

Proof. A K-group solution costs $(K-1)S_{\text{stop}} + c_{\text{switch}} \cdot z_K$. It beats the K^*-group optimal iff $c_{\text{switch}}(z_{K^*} - z_K) > (K - K^*)S_{\text{stop}}$, i.e. $\rho < \Delta(I, K)/(K - K^*)$. Collapse fails iff some K satisfies this; collapse holds iff $\rho \geq \rho_c(I)$. □ □

Remark 6.4 (The 6-job ring)**.** For the ring instance (Part V, Example 1): $K^* = 3$, best 3-group GSP cost $z_3 = 4$, SSP optimum $z_4 = 3$ with 4 groups. So $\rho_c(\text{ring}) = \Delta(I, 4)/(4-3) = (4-3)/1 = 1$. For $\rho > 1$: JGP is optimal (TSIP regime). For $\rho \leq 1$: 4 configurations may be beneficial.

Summary

Variant	Cost f	Collapse?	Status
U-SSP	$\lvert C_{k+1} \setminus C_k \rvert$	Not in general	Counterex. in Part V
TSIP	$\mathbf{1}[C_k \neq C_{k+1}]$	Yes, always	**Proved**
f_S, $S_{\text{stop}} > nbc$	Fixed stop + per-tool	Yes	**Proved**
TD-SSP, $S_{\text{stop}} \gg c_t$	Fixed stop + tool-dep.	Lexicographic	**Proved**
f with C1+C2+C3	General	Yes	**Proved (sufficient)**

References

[Adjiashvili et al.(2015)Adjiashvili, Bosio, and Zemmer] D. Adjiashvili, S. Bosio, and K. Zemmer. Minimizing the number of switch instances on a flexible machine in polynomial time. *Operations Research Letters*, 43(3):317–322, 2015.

[Furrer and Mütze(2017)] M. Furrer and T. Mütze. An algorithmic framework for tool switching problems with multiple objectives. *European Journal of Operational Research*, 259(3):1003–1016, 2017.

[Privault and Finke(1995)] C. Privault and G. Finke. Modelling a tool switching problem on a single nc-machine. *Journal of Intelligent Manufacturing*, 6(2):87–94, 1995.